



INVESTIGATING AI: A PRIMER FOR DECISION-MAKERS

INTRODUCTION

We live in a world where AI is everywhere. Banks use it to approve or reject loan applications. Companies use it to shortlist job candidates. Hospitals use it to assist with diagnoses. Fraud detection, customer service, financial forecasting - AI is behind all of it. Most organizations simply trust that these systems are working correctly. But what if they are not? What if the AI itself has been interfered with?

That is exactly what Machine Learning Forensics deals with. It is about investigating the AI itself when something has gone wrong. This primer looks at three specific ways AI systems can be attacked, and explains how investigators go about tracing each one.

Machine Learning Forensics (ML Forensics) is the discipline of investigating AI systems themselves, detecting when they have been tampered with, manipulated, or abused, and tracing exactly how it happened. This is distinct from simply using AI to catch criminals. Where traditional cybersecurity uses AI as a tool to detect threats, ML Forensics turns the investigation inward: it asks whether the AI tool itself has been compromised.

This matters because AI systems are now making real decisions in organizations - screening job applicants, detecting fraud, diagnosing patients. If the AI making those decisions has been corrupted, the damage can be enormous and completely invisible.

Training Data Poisoning - Corrupted at Birth

Every AI system learns by studying examples. A fraud detection AI studies past fraud cases. A medical diagnostic AI studies thousands of clinical records. This collection of examples is called training data and it is the foundation of everything the AI will ever do.

Training data poisoning is when an attacker secretly introduces false or misleading examples into that foundation. The AI then learns the wrong patterns from the very beginning. Its decisions become persistently flawed.

Once training data is poisoned, it opens the door to deeper attacks: inversion attacks, where sensitive information is extracted from the model, and backdoor attacks, where a hidden trigger is planted that causes the AI to behave maliciously on command.

A 2024 Nature Medicine study demonstrated just how fragile these systems can be: corrupting just **0.001%** of an AI's training data was enough to cause harmful errors, while the system continued passing every standard performance test.

How Investigators Trace Training Data Poisoning:

The first thing they do is watch how the AI behaves over time. If the AI's answers, reasoning, or accuracy suddenly change but no official update was made, that is a red flag.

Then they look at how it fails. A normal AI makes occasional random mistakes. A poisoned one fails in a pattern, it keeps getting things wrong in one specific area while doing fine everywhere else.

They also watch out for what researchers call hallucinations; where the AI confidently gives a completely wrong answer as if it were fact. This usually means it was trained on false information that it picked up as truth.

After that, investigators go deeper into the actual training data and run statistical checks, scanning thousands of data entries to find the ones that do not belong. The entries that look out of place, inconsistent, or statistically unusual are the ones that get flagged.

Investigators also want to know when the bad data got in, which specific entries caused the problem, and who had access to the data pipeline at that time.

Model Manipulation:

Model manipulation is when someone secretly goes into the internal files of an AI system and changes how it works from the inside. This is different from an adversarial attack, where someone tries to confuse the AI by feeding it tricky inputs. With model tampering, the attacker does not try to trick the AI, they reprogram it directly.

How Investigators Trace Model Manipulation:

The first tool they use is a cryptographic hash; a unique digital fingerprint of the model generated at the moment it was deployed. If anything inside the model changes, even slightly, the fingerprint changes too. Investigators compare the model's current fingerprint against the original stored copy. A mismatch is immediate evidence that something was altered.

They also examine the model's weights - the internal numbers that determine how the AI thinks and responds. Investigators compare current weights against verified earlier versions, looking for changes that were never part of an authorized update.

Investigators also consider data provenance which is tracing the full history of the model to understand where it came from, who built it, who maintained it, and who had the ability to modify it.

The business consequences of undetected model manipulation are serious. A tampered fraud detection model could quietly stop flagging certain transactions. A tampered hiring AI could systematically favor specific candidates. Beyond the direct damage, organizations also face data breaches, fraudulent decision-making liability, and severe regulatory penalties, particularly under frameworks like the EU AI Act, where demonstrating model integrity is a legal requirement.

For clarity sake, Data Poisoning differs from Model Manipulation

In **Training Data Poisoning**, the attack happens **before** the AI is built.

Someone corrupts the information the AI learns from. So the AI grows up learning the wrong things.

In **Model Tampering**, the attack happens **after** the AI is already built and deployed.

The AI was built correctly and was working fine. Then someone sneaks in and secretly changes its internal settings. Now it behaves differently, but nobody authorized that change.

Synthetic Data Abuse:

Synthetic data is any content that was **generated by AI** rather than recorded or written by a real human. This includes:

- Fake videos of real people saying things they never said (deepfakes)
- AI-generated voice recordings that sound exactly like a specific person
- Fake images of people that do not exist
- AI-written text pretending to be from a real person

Synthetic data itself is not always bad and companies use it legitimately for testing and research. The abuse happens when it is used to **deceive, defraud, or manipulate**.

The Arup Case: Hong Kong, January 2024

A finance employee at Arup, the British engineering company, received an invitation to a video call. On that call he saw the company's CFO and several senior colleagues. The faces were familiar. The voices sounded exactly right. They told him the company needed urgent fund transfers.

He transferred **\$25.6 million** across 15 transactions.

Every single person on that call except him was an AI-generated deepfake. Nobody else was real. Arup later confirmed: *"Fake voices and images were used."*

How Investigators Trace Synthetic Data Abuse

AI-generated video contains tiny imperfections that the human eye misses but forensic tools catch. Investigators look for:

- Unnatural blinking: deepfake faces often blink too rarely or in an unnatural rhythm
- Blurry edges around the hairline or ears where the AI struggled to render detail
- Lip movements that are slightly out of sync with the audio

AI-generated voices leave traces in the sound itself. Investigators analyze the audio waveform and look for:

- Unnatural breathing patterns: real humans breathe, AI voices often do not
- Frequency anomalies: certain sound frequencies that appear in AI-generated audio but not in natural human speech
- Robotic smoothness

Also, Every genuine video or audio file carries metadata which are hidden information about where it was recorded, what device was used, what software created it, and when. AI-generated content either has no metadata or has metadata that does not match the claimed source.

Investigators also look at the human side. Things like: “did anyone verify the communication through a separate channel? Did anyone call back on a known number? Was there a follow-up email from a verified address?” are verified.

Why Executives Should Care

- The technology to create a convincing deepfake is now **cheap and widely available**
- You cannot trust your eyes or ears anymore so therefore verification must be built into organizational processes

CONCLUSION

The three attacks covered in this primer: training data poisoning, model manipulation, and synthetic data abuse are all built around the same idea: stay hidden for as long as possible. A poisoned AI still passes its tests. A tampered model still looks like the original. A deepfake still sounds and looks like a real person. The damage happens quietly, and by the time it is noticed, it has usually already been done. Machine Learning Forensics exists because traditional cybersecurity was not built to catch this kind of threat. It looks past the surface and the normal-looking outputs and investigates the AI itself.